

Michael Eyram Alomatu

Full Stack Software Engineer · CS & Engineering Student

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[LinkedIn Profile](#) · [GitHub Profile](#)

PROFILE

Full Stack Software Engineer with hands-on experience building data-driven web applications across technical and operational domains. Proficient in HTML5, CSS3, and Vanilla JavaScript, with working knowledge of React, Python, Django, and SQL across both frontend and backend development. I integrate AI tooling as part of my development workflow to accelerate delivery and produce efficient, reliable software, while retaining full ownership of all architectural and technical decisions. Practical experience includes contributing to enterprise systems during internships at the Ghana Grid Company (GRIDCo) and the Ghana Ports and Harbours Authority, with a consistent focus on building software that solves real, specific problems.

EDUCATION & CERTIFICATIONS

Bachelor of Science — Computer Science and Engineering

Jan 2023 – Present

University of Mines and Technology (UMaT) · Tarkwa, Ghana

ALX AI Career Essentials Certification

May – Jun 2024

ALX Africa · Online

West African Senior Secondary Certificate (WASSCE)

Sep 2017 – Sep 2020

Opoku Ware School · Kumasi, Ghana

EXPERIENCE

Technical Intern — Ghana Grid Company (GRIDCo)

Nov 2024 – Dec 2024

IT Department · Business Applications Unit · Tema, Ghana

- Contributed to the development of an internal Asset Management System, applying Object-Oriented Programming (OOP) principles in a production codebase
- Supported business application workflows by aligning technical solutions with operational requirements in a mission-critical infrastructure environment
- Participated in team-based development and collaborative problem-solving within a high-availability national electricity transmission setting
- Gained practical exposure to cybersecurity awareness and best practices within critical national infrastructure systems

Technical Intern — Ghana Ports and Harbours Authority

Oct 2023 – Dec 2023

Port of Takoradi · IT Department · Takoradi, Ghana

- Collaborated in a Scrum-based team environment, participating in regular stand-up meetings and sprint planning activities
- Contributed to the development of a web application built with Python and Odoo to enhance inter-departmental communication across operational units
- Supported the translation of operational requirements into functional software features within an enterprise system
- Applied Agile methodologies in a real-world, infrastructure-driven development environment

Multimedia Personnel — ICGC

Jan 2023 – Present

Part-time · Volunteer

- Managed audiovisual production for live events including professional videography and multimedia content
- Maintained and operated technical equipment to ensure seamless event presentations

PROJECTS

[Advanced Bearing Capacity Analyzer](#) · HTML5 · Vanilla CSS3 · Vanilla JavaScript · HTML5 Canvas

Browser-based geotechnical engineering tool implementing Terzaghi's Bearing Capacity Theory. Architected task decomposition, validated all mathematical logic, and built a live HTML5 Canvas foundation diagram. Calculates foundation load capacity and evaluates design safety factors handling groundwater depth and foundation geometry — zero dependencies, fully browser-native.

[Soil Compaction Analyzer](#) · HTML5 · CSS3 · Vanilla JavaScript · Chart.js

Decision-support tool for geotechnical engineers developed using an agentic iteration workflow with rigorous human-in-the-loop review. Processes raw Proctor test data to calculate Maximum Dry Density (MDD) and renders interactive compaction curves for direct field use. Offline-capable.

[Water Chlorination & Dosage Dashboard](#) · HTML5 · Vanilla CSS3 · Vanilla JavaScript · Chart.js

Operational web dashboard for water treatment operators integrating flow rate inputs with real-time chlorine dosage calculations and dynamic data visualization. Managed full technical architecture and UI validation. Client-side only — fully offline once loaded, no backend required.

SKILLS

Languages: Python · JavaScript · HTML/CSS · SQL · C#

Frameworks & Libraries: React · Django · Streamlit · Pandas · Chart.js · Odoo

Tools & Platforms: Google Antigravity · Firebase · Git · GitHub

Core Competencies: Agentic AI Orchestration · Technical Architecture · Web Development · Rapid Prototyping · Agile / Scrum

Soft Skills: Effective Communication · Adaptability · Critical Thinking · Problem-Solving · Execution-Oriented · Collaborative Team Player

EXTRACURRICULAR ACTIVITIES

- Master of Ceremonies (MC) — Association of Computer Science and Engineering Students (ACSES), University of Mines and Technology, Tarkwa. Regularly serves as MC for departmental meetings and events, managing programmed flow, facilitating structured discussions, and ensuring professional proceedings for the student body.
- Programmer & Member, Robotics Club — Opoku Ware School (Jul 2018 – Sep 2020). Developed algorithms for line-following and obstacle avoidance; collaborated on robotic design and testing.
- Contestant, The Sharks' Quiz — Opoku Ware School
- Contestant, Ultimate High School Quiz Competition — Opoku Ware School